

JunaidTeam @ HASTIKA 2026

Hate Speech and Target Category Identification in Kannada-English Code-Mixed Text

Team Members: Junaid Abedin Rafi, Rifah Tasnim Jui, Tasfia Tabassum Toma

CodaBench Team: JunaidTeam (username: junaidabedinrafi)

Shared Task: HASTIKA @ ICON-2026 (CodaBench Competition #17784)

1. Introduction

Social media platforms in multilingual regions such as India frequently contain code-mixed text, where a native language is blended with English within the same utterance. This phenomenon, often called Kanglish when Kannada is involved, poses significant challenges for automatic hate speech detection due to transliteration inconsistencies, spelling variation, and the scarcity of annotated resources for low-resource language pairs. The HASTIKA shared task at ICON-2026 addresses this gap using a corpus of 8,058 manually annotated Kannada-English code-mixed YouTube comments. The task is split into two independent sub-tasks:

- **Task A - Binary Hate Speech Detection:** classify each comment as *Hate* or *Non-Hate*.
- **Task B - Fine-Grained Hate Speech Classification:** classify each hate comment into one of six target categories: Gender, Political, Religion, Geo-political, Violence, Others.

This report documents our approach, dataset analysis, methodology, results, and possible future improvements for both sub-tasks.

2. Dataset Analysis

The HASTIKA corpus contains 8,058 Kannada-English code-mixed YouTube comments drawn from politics, religion, entertainment, sports, and social-issue discussions, manually annotated by language experts. The data was released in two phases: a labeled training/validation split provided at task launch, and an unlabeled test split released later for final evaluation (gold labels withheld by organizers).

2.1 Task A - Binary Hate Speech Detection

Label	Train	Test	Total
Hate	3,160	~395	3,950
Non-Hate	3,286	~411	4,108
Total	6,446	806	8,058

The training split is nearly balanced between Hate (49.0%) and Non-Hate (51.0%) classes, which reduces the risk of severe majority-class bias for Task A.

2.2 Task B - Fine-Grained Classification (hate comments only)

Category	Train	Test	Total
Gender	1,362	~171	1,703
Political	559	~70	699
Others	449	~56	561
Religion	382	~48	478
Violence	221	~28	277
Geo-political	186	~23	232
Total	3,159	396	3,950

Task B exhibits substantial class imbalance: the Gender category alone accounts for 43.1% of all hate comments, while Geo-political (5.9%) and Violence (7.0%) are heavily underrepresented. This imbalance motivated our choice of a class-weighted classifier and the use of Macro-F1 (rather than Accuracy) as the primary evaluation metric, since Macro-F1 weights all six categories equally regardless of frequency.

2.3 Text Characteristics

Qualitative inspection of the comments shows heavy code-mixing (Kannada transliterated into Roman script interleaved with English), inconsistent spelling of the same Kannada words across comments, stray HTML artifacts (e.g. `<br>`, `"`), emoji, and occasional non-ASCII mojibake characters from encoding issues in the source YouTube comments. These characteristics directly informed our preprocessing and feature-engineering choices described below.

3. Methodology

3.1 Preprocessing

- Removal of stray HTML tags (e.g. `<br>`) and HTML entities (`"`, `&`).
- Removal of URLs.
- Whitespace normalization.
- Deliberately **no** aggressive normalization of spelling variants: since the text is transliterated Kanglish, non-standard spelling is itself informative signal rather than noise, so we avoided stemming or spelling correction that could destroy this signal.

3.2 Feature Extraction

We used TF-IDF vectorization combining two complementary views of the text:

- **Word-level n-grams (1-2):** captures lexical/phrase-level signal (e.g. frequent slurs or political terms).
- **Character n-grams (2-5, word-boundary aware):** critical for code-mixed, transliterated text, since the same Kannada word can be spelled multiple different ways across comments (e.g. 'nimge' / 'nimage'). Character n-grams allow the model to generalize across such spelling variants by sharing sub-word features.

The two feature sets were concatenated (horizontally stacked) into a single sparse feature matrix before being passed to the classifier.

3.3 Model

We used **Logistic Regression** (scikit-learn, L2-regularized, C=5, max_iter=2000) as our classifier for both sub-tasks, with **class_weight='balanced'** to counteract class imbalance (particularly important for Task B). Logistic Regression was chosen as a strong, fast, and well-understood baseline for high-dimensional sparse TF-IDF features, well suited to the short development timeline of the shared task.

3.4 Validation Strategy

An 85/15 stratified train/validation split was carved out of the labeled training data to estimate Macro-F1 before generating official predictions. Once validated, the final model for each task was retrained on 100% of the available labeled training data before predicting on the official (unlabeled) validation/test inputs, to make maximal use of the limited training signal.

3.5 Pipeline Summary

Raw comment → text cleaning → TF-IDF (word + char n-grams) → class-weighted Logistic Regression → predicted label → predictions.csv → CodaBench / Google Form submission.

4. Results

Table 3 reports both our internal held-out validation performance and the official scores obtained on CodaBench's Development-phase leaderboard (evaluated on the provided validation_inputs files).

Task	Internal Macro-F1	Internal Accuracy	Leaderboard Macro-F1	Leaderboard Accuracy
Task A - Binary	0.82	0.82	0.80	0.80
Task B - Fine-Grained	0.61	0.70	0.60	0.71

On the public Development leaderboard (39 participating teams as of the submission deadline), our Task A score of 0.80 Macro-F1 placed in the upper-middle of the field (top score: 0.90), while our Task B score of 0.60 Macro-F1 was competitive among teams that attempted the harder six-way classification sub-task.

As expected, Task B (six-way, imbalanced) scored substantially lower than Task A (balanced binary), confirming that the rare categories (Geo-political, Violence) are the primary bottleneck for overall Macro-F1 on this sub-task.

5. Discussion / Error Analysis

- **Class imbalance in Task B** is the single biggest limiting factor: categories like Geo-political (186 train examples) and Violence (221 train examples) have roughly 7x fewer examples than Gender (1,362), making it difficult for the model to learn reliable decision boundaries for these classes even with class weighting.

- **Spelling variation** in transliterated Kannada means the same underlying word can appear in many surface forms; while character n-grams partially mitigate this, out-of-vocabulary variants not seen in training still hurt recall.
- **Category overlap**: some comments plausibly span multiple target categories (e.g. a politically-motivated comment that is also gendered), but the task requires a single label, which introduces some irreducible label noise.

6. Future Work

- Fine-tune multilingual transformer models pretrained on Indian languages (e.g. MuRIL, IndicBERT) or general multilingual transformers (XLM-RoBERTa, mBERT), which can capture contextual and sub-word semantics beyond what TF-IDF offers.
- Use Focal Loss or targeted oversampling/data augmentation specifically for the minority Task B categories (Geo-political, Violence) to improve their recall.
- Explore ensembling multiple models (e.g. TF-IDF + Logistic Regression combined with a transformer) via soft-voting to combine the strengths of lexical and contextual representations.
- Perform systematic error analysis using a confusion matrix on held-out data to identify which category pairs are most frequently confused.

7. Conclusion

We developed a TF-IDF + class-weighted Logistic Regression baseline for the HASTIKA @ ICON-2026 shared task, achieving 0.80 Macro-F1 on Task A (Binary Hate Speech Detection) and 0.60 Macro-F1 on Task B (Fine-Grained Classification) on the Development leaderboard. The approach is lightweight, fast to train, and provides a solid baseline; the discussion above outlines concrete directions (transformer fine-tuning, minority-class handling) for improving performance further in future iterations.

8. Repository and Submission Links

- Code repository (GitHub): to be added after upload.
- Shared task CodaBench page: <https://www.codabench.org/competitions/17784/>
- Official starting kit: <https://github.com/shankarb14/Hastika-ICON2026>

Citation: Kavatagi, S., Rachh, R. (2025). HASTIKA: hate speech and target identification in Kannada-English code-mixed text. *Language Resources and Evaluation*, 59(3), 2811-2856.